Benchmarks Are Engines, Not Cameras

How ML evaluation tools reshape the knowledge they measure

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Overview

Part 1

Benchmarks are engines, not cameras

The Black-Scholes story

In 1973, Fischer Black and Myron Scholes published a formula for pricing options — the Black-Scholes model.

Before it, options pricing was mostly intuition and guesswork. Different traders quoted wildly different prices for the same contract.

The formula said: given a stock price, volatility, time to expiry, and interest rate, there is **one correct price** for every option.

$$C = S\,\Phi(d_1) - K\,e^{-rT}\Phi(d_2)$$

The same year, the Chicago Board Options Exchange opened. Traders carried printouts of Black-Scholes pricing tables onto the trading floor.

The model made itself true

At first, actual option prices **deviated** from Black-Scholes predictions.

Then something strange happened. Over the next decade, market prices **converged** toward the formula's predictions.

The formula didn't *describe* a pre-existing reality. Traders using the formula **created the reality it described**. And when the assumptions broke in 1987, prices diverged just as fast.

Camera vs. engine

The sociologist Donald MacKenzie studied this episode in *An Engine, Not a Camera* (2006) and asked: if a model **changes** the thing it measures, is it still a **measurement**?

The answer: no. The model is not a **camera** that passively records what's there. It's an **engine** that reshapes the world.

Camera

Passively records what's there

↓

measurement

vs.

Engine

Reshapes the thing it measures

↓

intervention

Our claim: benchmarks do the same thing

We argue that **ML benchmarks** operate exactly like Black-Scholes:

1. A benchmark is introduced to **measure** model capability.
2. Researchers optimize against it. Labs orient entire research programs around it.
3. The benchmark stops measuring a pre-existing ability — it starts **shaping** what gets built.
4. Eventually the benchmark saturates, and the community recognizes it was measuring the wrong thing all along.

The benchmark is not a camera. It's an engine.

We will now explore what that means — for statistical validity, for proof standards in MI, and for peer review.

Three levels of performativity

MacKenzie distinguishes how *deeply* a model can reshape the world:

Generic: Practitioners use the model in their work

Effective: Using the model *changes* the phenomenon it describes

Barnesian: The world conforms to the model — the map redraws the territory

stronger

Plus a failure mode: **counterperformativity** — when widespread adoption of the model causes the world to *diverge* from its predictions.

Concrete examples of each level

Level	Finance	ML
Generic	Traders use Black-Scholes to price options	Researchers use ImageNet to evaluate models
Effective	Using the formula changes how options are priced	Optimising for ImageNet changes what architectures get built
Barnesian	Market prices converge to the formula's predictions	"State of the art" = "best ImageNet score"
Counter-	Gaussian copula creates the correlated risk it assumed away	ML evaluation inflates the false positives it aims to prevent

The same instrument can be at different levels at different times. Black-Scholes was Barnesian in the 1980s and counterperformative in 1987.

Counterperformativity: the Gaussian copula

The clearest example of a model **destroying** what it assumed:

1. David Li (2000) publishes the **Gaussian copula** formula for pricing mortgage-backed securities. Core assumption: defaults are only weakly correlated.
2. Every major bank adopts it. Trillions of dollars of CDOs are priced using this one model.
3. Because everyone uses the same model with the same assumption, they all hold the same risks. The **actual** correlation between defaults increases.
4. In 2007–08, correlated defaults cascade through the system. The model's assumption of low correlation **created** the high correlation that broke it.

The safety model generated the danger it was supposed to prevent. We argue the same structure operates in ML benchmarks.

Mapping this to ML benchmarks

Financial models

Black-Scholes formula

↓

Trading desks adopt it

↓

Prices conform to model

↓

Model “made itself true”

ML benchmarks

ImageNet introduced

↓

Labs optimize against it

↓

Score = “visual understanding”

↓

Overfitting the benchmark can’t detect

The channels are different (trading desks vs. conference leaderboards), but the dynamic is the same: the instrument reshapes what it measures.

Where the analogy breaks (and where it holds)

Where it holds

- The instrument reshapes the thing it claims to measure
- Adoption is driven by institutional incentives, not just technical merit
- The community treats the model's output as ground truth
- Failure modes emerge from collective adoption, not individual misuse

Where it differs

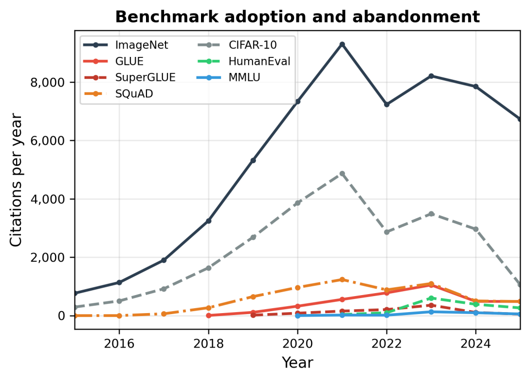
- No single formula — many benchmarks, loosely coordinated
- No direct financial feedback loop (prices don't clear)
- We claim **effective** performativity, not Barnesian — benchmarks reshape research priorities, not reality itself
- The "crash" is slow replacement, not a sudden crisis

We're not claiming ML = finance. We're claiming the **structure** is the same: an instrument presented as a camera that functions as an engine.

The benchmark lifecycle

Every benchmark goes through four phases:

1. **Camera.** Introduced as a measurement. Passively records model performance.
2. **Adoption.** Researchers optimize against it. Leaderboards emerge. Score = the capability.
3. **Saturation.** Optimisation changes what models are built. The benchmark reshapes training, data, architecture.
4. **Crash.** Benchmark is saturated or recognized as measuring the wrong thing. Replaced. Cycle restarts.



NLP benchmarks (GLUE, SuperGLUE) crash in 3–5 years.
Vision benchmarks (ImageNet) persist as cultural institutions.

NLP benchmarks crash fast

Benchmark	Domain	Introduced	Peak cites/yr	Decline to 2024
GLUE	NLP	2018	1,052	−54%
SuperGLUE	NLP	2019	359	−69%
SQuAD	NLP	2016	1,237	−59%
ImageNet	Vision	2009	9,305	−16%
CIFAR-10	Vision	2009	4,870	−39%

Key pattern: NLP benchmarks are abandoned when models saturate them — not when the underlying capability is "solved."

GLUE wasn't replaced because language understanding was solved. It was replaced because BERT-scale models maxed out the score and the community needed something harder.

The benchmark tracks **social utility** — the ability to rank models.

Part 2

Proof cultures and the denominator problem

What is mechanistic interpretability?

The goal: understand **how** neural networks compute, not just *what* they compute.

Standard ML evaluation

- Run model on test set
- Report accuracy, F1, BLEU, etc.
- The model is a **black box**

This tells you *what* the model does, but not *why* or *how*.

Mechanistic interpretability

- Open up the model
- Identify internal components (features, circuits, causal variables)
- Explain **how it works**

This is a young field. There is not yet broad agreement on what counts as evidence.

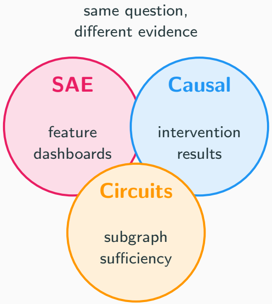
That disagreement is the subject of Part 2.

What is a "proof culture"?

MacKenzie studied formal verification: what counts as a valid "proof" that a computer system is correct.

He found that **logicians**, **mathematicians**, and **engineers** accepted different kinds of arguments as proof. The boundary was shaped by funding and institutional authority, not just by logic.

The same thing is happening in mechanistic interpretability. Three communities are trying to answer the same question — "how do neural networks compute?" — using **incompatible evidence standards**.



Three communities, three evidence standards

Sparse Autoencoders

Feature dashboard

"Golden Gate Bridge"

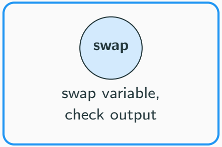
activation examples

steering demo

Valid if features are **interpretable** and monosemantic.

Causal Abstraction

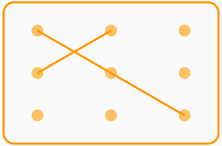
Intervention result



Valid if **interventions** produce predicted effects.

Circuit Discovery

Circuit subgraph



Valid if the circuit is **necessary and sufficient** for the behavior.

These are not interchangeable. A feature dashboard would not satisfy a causal abstraction reviewer; an intervention result would not satisfy a circuits reviewer. Each community has its own "what counts as evidence."

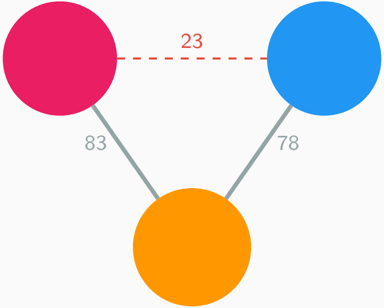
Subfields don't cite each other

We analyzed citation links among 688 MI papers (209 SAE, 40 Causal, 128 Circuits, 311 general).

From \ To	SAE	Causal	Circuits
SAE	326	**14**	45
Causal	**9**	63	16
Circuits	38	62	157

SAE $\leftrightarrow$ Causal: **23 cross-citations** across 16,720 possible directed pairs.

That's **0.14% density**. Both communities cite circuits literature, but barely cite each other.



Same foundation, no dialogue

Citation insularity: permutation test

Is the low cross-citation density just noise?

We shuffled community labels 10,000 times and recomputed cross-density each time.

Metric	**Value**
Observed cross-density	0.21%
Null (shuffled) mean	0.26%
Null 95% range	[0.23%, 0.28%]
Within-community density	0.35%
Within/cross ratio	1.6×
Permutation p-value	$p = 0.0002$

Community labels predict citation structure.

The insularity is **real**, not an artifact of small samples.

Sensitivity to corpus boundary:

Subset	n	**Ratio**
Seed papers	18	2.4×
Full corpus	688	1.6×
Top-cited (25%)	173	0.9×

The most-cited papers **do** bridge communities (ratio ≈ 1). Fragmentation lives in the long tail — the average researcher stays in their lane.

Fragmentation is increasing

The within/cross citation density ratio by year shows communities becoming **more insular** over time, not less.

Year	Papers	Edges	Cross%	Ratio
2021	6	5	60%	1.0×
2022	11	10	70%	1.1×
2023	74	108	75%	0.5×
2024	113	154	65%	1.0×
2025	**224**	**280**	**51%**	**2.3**×
2026	360	643	50%	2.0×

Dashed line = no insularity. As the field grows (especially SAE work post-2023), communities cite inward more.

Author overlap: separate tribes

Only **4.3%** of 724 unique authors publish in more than one MI community.

Pair	Shared	Jaccard
SAE ↔ Causal	0	0.000
SAE ↔ Circuits	6	0.015
Causal ↔ Circuits	1	0.006

SAE and causal abstraction share **zero authors**. The communities are not just reading different papers—they are **different people**.

Bridge authors (2+ communities):

Author	Communities
Arthur Conmy	circuits, sae
Neel Nanda	circuits, sae
Chris Olah	circuits, sae
Yonatan Belinkov	circuits, causal
Atticus Geiger	causal, general
Samuel Marks	circuits, general
David Bau	circuits, general
Aaron Mueller	circuits, general

Bridges connect circuits↔SAE or circuits↔general.

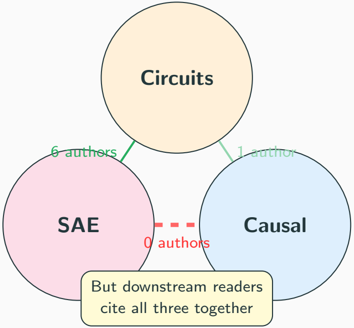
Nobody bridges SAE↔causal.

Co-citation: readers see one field, authors see three

Despite low direct citation, **downstream papers cite seeds from different communities together**.

Cross-community pair	Co-cited by
Geiger (causal) + Wang (circuits)	180 papers
Marks (circuits) + Cunningham (SAE)	172 papers
Wang (circuits) + Cunningham (SAE)	163 papers
Meng (circuits) + Cunningham (SAE)	138 papers
Geiger (causal) + Conmy (circuits)	131 papers

76 cross-community pairs are co-cited, with 2,448 total cross co-citations.



The reader community sees MI as one field. The producing communities do not.

Shared ancestors, divergent reading

Seed papers across all three communities share **16 common ancestors**—foundational papers cited by seeds in all three communities.

Papers cited by all 3 communities
Toy Models of Superposition
Zoom In: An Introduction to Circuits
Language Models are Unsupervised Multitask Learners
Axiomatic Attribution for Deep Networks
Finding Neurons in a Haystack
How does GPT-2 compute greater-than?
Pythia: A Suite for Analyzing LLMs
Investigating Gender Bias via Causal Mediation

Pairwise Jaccard similarity of reference sets:

Pair	Jaccard
Causal $\leftrightarrow$ Circuits	0.119
Circuits $\leftrightarrow$ SAE	0.086
Causal $\leftrightarrow$ SAE	0.084

Low Jaccard similarity confirms divergent reading. The communities share intellectual roots but have **branched** — reading the same foundations without reading each other's extensions.

59 additional papers are shared by exactly 2 communities.

The denominator problem

An SAE trains a dictionary of N features. The researcher analyzes them and ultimately chooses to display the k most interesting ones.

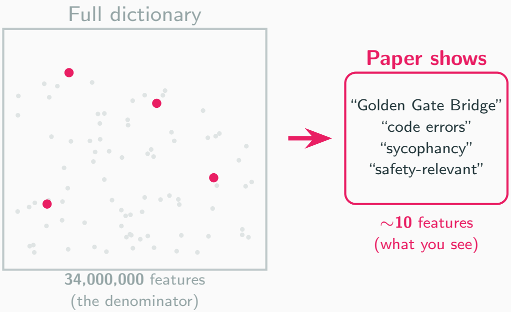
This is a multiple hypothesis test with $m \geq N$. None of the papers we audited corrects for it.

The worked example:

Templeton et al. (2024) trained **34 million** features on Claude 3 Sonnet. They showed $\sim$10 in detail.

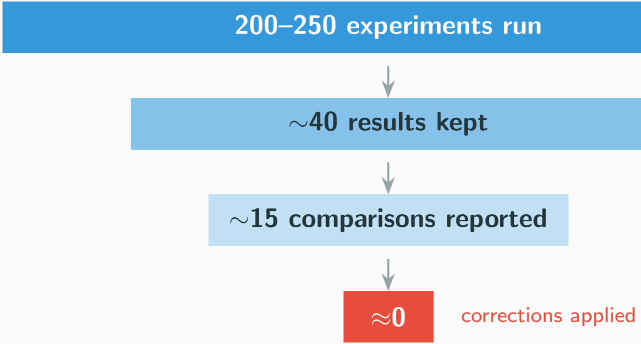
Selection ratio: **3.4 million to one**.

If even 0.01% of features are spuriously interpretable, that's **3,400 false positives** — dwarfing the 10 shown.



The practitioner's funnel

Even outside MI, the denominator is invisible. A typical ML project:



The paper reports 15 comparisons. The reviewer sees 15. The reader sees 15. But 200+ were run. **The denominator is invisible by default.**

The multiple comparisons problem

A simple thought experiment:

Flip a fair coin 20 times. The chance of getting heads at least once is not 5% — it's **64**%.

Now imagine running 20 experiments, each with a 5% false positive rate. The chance that *at least one* gives a false positive is also 64%.

This is well understood in medicine and genetics. Researchers routinely correct for it. In ML, almost nobody does.

Metric 1: Family-wise error rate (FWER)

The probability that *at least one* of your comparisons is a false positive.

$$\mathrm{FWER} = 1 - (1 - \alpha)^m$$

At $\alpha = 0.05$:

Comparisons (m)	**FWER**	
5	23%	(typical single experiment)
20	64%	(typical paper)
100	99.4%	(hyperparameter search)
1,000	≈100%	(architecture search)

With enough comparisons, finding "significant" results is guaranteed by chance alone.

Controlling FWER means ensuring this probability stays below α.

Metric 2: False discovery rate (FDR)

The expected *proportion* of false positives among your claimed discoveries.

$$\text{FDR} = \mathbb{E}\left[\frac{\text{false positives}}{\text{total rejections}}\right]$$

At $q = 0.05$, no matter how many tests you run:

Tests	**Discoveries**	**Expected false**	
20	4	$\sim$0.2	(5% of 4)
100	20	$\sim$1	(5% of 20)
1,000	50	$\sim$2.5	(5% of 50)
10,000	200	$\sim$10	(5% of 200)

FDR scales with discoveries, not total tests. It's less strict than FWER — you accept some false positives — but you find far more real effects. The tradeoff: you know $\sim$5% of your results are wrong, but not *which* 5%.

Control 1: Bonferroni correction (1936)

The simplest fix has existed since 1936:

If you run m comparisons, divide your significance threshold by m.

$$\alpha_{\text{corrected}} = \frac{\alpha}{m} = \frac{0.05}{m}$$

Field	**Typical m**	**Corrected α**	**Correction used?**
Clinical trials	3–5	0.01	Always (FDA requires it)
Genomics (GWAS)	1,000,000	5×10^{-8}	Always
Psychology	5–20	0.003–0.01	Usually
ML (standard)	100–1,000+	would be 10^{-4}	Almost never
ML (MI/SAEs)	1,000–34,000,000	would be 10^{-9}	Never

Control 2: Holm–Bonferroni method (1979)

A strictly more powerful version of Bonferroni:

1. Sort your m p-values from smallest to largest: $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$
2. Compare each $p_{(i)}$ against a *different* threshold: $\frac{\alpha}{m-i+1}$
3. Reject from the smallest up, stop at the first non-rejection

Rank	**P-value**	**Bonferroni threshold**	**Holm threshold**
1st	0.003	$0.05/5 = 0.010$	$0.05/5 = 0.010$
2nd	0.009	$0.05/5 = 0.010$	$0.05/4 = 0.0125$
3rd	0.02	$0.05/5 = 0.010$	$0.05/3 = 0.017$

Holm rejects *at least* as many hypotheses as Bonferroni. It is uniformly more powerful while still controlling FWER. There is no reason to use Bonferroni over Holm.

Control 3: Benjamini–Hochberg procedure (1995)

A different philosophy: control the **false discovery rate** (FDR) instead of FWER.

FWER: probability of *any* false positive $\leq \alpha$

FDR: expected *proportion* of false positives among rejections $\leq q$

1. Sort p-values: $p_{(1)} \leq \cdots \leq p_{(m)}$
2. Find the largest k such that $p_{(k)} \leq \frac{k}{m} \cdot q$
3. Reject all hypotheses $1, \ldots, k$

This is much less conservative than Bonferroni/Holm. If you run 1,000 tests at $q = 0.05$, you accept that $\sim$50 of your "discoveries" may be false — but you find far more real effects.

Widely used in genomics (thousands of genes tested simultaneously). Almost never used in ML.

Which correction to use?

Method	Controls	Best for
Bonferroni	FWER (strictest)	Very few tests; need zero false positives (clinical trials)
Holm	FWER (uniformly better)	Same situations as Bonferroni — strictly dominates it
Benjamini–Hochberg	FDR (more lenient)	Many tests; some false positives acceptable (screening, genomics)

For ML experiments (5–20 reported comparisons): Holm is the obvious default. It costs nothing, requires no assumptions beyond independence, and is trivial to implement.

For MI/SAE feature selection (thousands–millions of candidates): BH-style FDR control is more appropriate, since FWER becomes impossibly strict at that scale.

What other fields require

Field	Requirement	Enforced by
Clinical trials	Pre-registered correction (usually Bonferroni or Hochberg)	FDA / EMA (law)
Genomics (GWAS)	Genome-wide threshold 5×10^{-8} or BH-FDR	Journal policy
Psychology	Correction mandatory since replication crisis	APA guidelines
Neuroscience	Cluster-level FWE or FDR for imaging	Field norm

These aren't suggestions — in clinical trials, submitting results without pre-specified multiplicity adjustment is grounds for regulatory rejection. The FDA has mandated this since 1998.

ML has no equivalent authority, no pre-registration norm, and no review checklist that asks "how many comparisons did you actually run?"

What counts as a "comparison" in ML?

This is the key question. In ML, comparisons are often **implicit**:

- Every hyperparameter setting you tried
- Every architecture variant you tested
- Every random seed you ran (and kept the best)
- Every baseline you compared against
- Every dataset split you evaluated on
- Every feature you looked at and decided to show

A paper reporting "5 experiments" may have actually run 200+. The **denominator** — the total number of comparisons — is invisible to the reader.

This is not fraud. It's the default workflow. But it has statistical consequences.

The scale of the problem in MI

Implicit comparison counts from real MI papers:

Paper	Comparisons (m)	FWER	Corrected?
Typical ML paper	100–500	>99%	No
IOI circuit (Wang+ 2023)	∼1,000	≈100%	No
SAEBench (Karvonen+ 2025)	∼10,000	≈100%	No
Scaling Monosemanticity (2024)	34,000,000	≈100%	No

Every one of these papers reports "interesting" features or circuits. At these scales, finding interesting-looking results is *guaranteed* — even in random noise.

The question is not whether SAE features look interpretable. It's whether the false positives are a problem.

Caveat: MI papers don't run classical hypothesis tests — there's no p-value to correct. But

How genomics solved this

Genome-wide association studies (GWAS) test $\sim$1,000,000 genetic variants per study. Same scale as MI. Their solution (standardized since $\sim$2005):

1. **Pre-register** the number of tests before looking at data
2. Apply BH-FDR at $q = 0.05$ for discovery (screening phase)
3. Replicate top hits in an **independent cohort** (validation phase)
4. Only replication survivors get published as “findings”

Result: genomics went from a replication crisis in the early 2000s to one of the most reliable fields in science. The false positive rate dropped from $>$50% to $<$5% within a decade.

MI could adopt the same pipeline: screen features/circuits with FDR control, then validate top candidates on held-out models or tasks. The infrastructure exists. The norm doesn’t.

What MI papers could actually do

Realistic corrections that cost almost nothing to implement:

1. **Report the denominator.** "We trained an SAE with 65k features. We inspected 200. We show 10." Just stating this changes the reader's interpretation.
2. **Apply BH-FDR to any quantitative screen.** If you rank features by a metric (e.g. activation sparsity, probe accuracy), apply BH at $q = 0.05$. One line of code: `statsmodels.stats.multitest.multipletests(pvals, method='fdr_bh')`
3. **Pre-register your hypothesis.** Decide what you're looking for *before* looking at the features. Even informally ("we predict layer 8 contains a gender direction") separates confirmation from exploration.

None of these require changing the research. They require changing the reporting.

Part 3

The counterperformativity of peer review

The paradox

Reviewers demand more experiments to **increase confidence**.

Each demanded experiment is a new comparison whose result is **selectively reported**.

The mechanism designed to reduce false positives may **structurally increase** them.

This is **counterperformativity**: the safety mechanism generates the risk it is designed to prevent. Same structure as the Gaussian copula in economics.

The data: 22,868 ICLR submissions

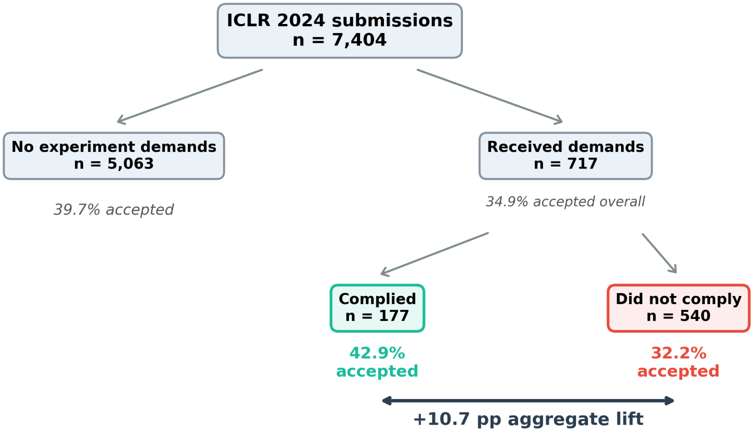
We analyzed every public submission to ICLR across three years:

Year	Papers	Demands	Complied	Comply acc.	Lift
ICLR 2025	11,672	1,797	336	50.6%	**+15.8 pp**
ICLR 2024	7,404	717	177	42.9%	**+10.7 pp**
ICLR 2023	3,792	357	78	47.4%	**+13.0 pp**

The aggregate compliance lift replicates across three years and is **growing**: $+13.0$ pp $\rightarrow$ $+10.7$ pp $\rightarrow$ $+15.8$ pp.

Sounds like the system works, right?

Demand compliance and acceptance




```
Simpson's paradox (score-stratified, n = 5,780):
Low (<4.5):  0.0% vs 0.0%    = 0.0 pp
Borderline:  7.5% vs 4.7%    = +2.9 pp
Mid (5.5-6.5): 50.0% vs 54.7% = -4.7 pp
High (>6.5):  94.9% vs 94.8%  = +0.1 pp
Aggregate lift is a composition effect.
```

What is Simpson's paradox?

A trend that appears in aggregate data **reverses** when you split the data into subgroups.

Classic example: Berkeley admissions

In 1973, UC Berkeley was sued for gender bias: overall, men were admitted at higher rates than women.

But when you looked *department by department*, women were admitted at **equal or higher** rates in most departments.

The aggregate gap existed because women applied to more competitive departments.

The lesson

An aggregate statistic can tell a completely different story from the stratified data.

This matters whenever subgroups have different sizes *and* different base rates.

We found exactly this pattern in ICLR compliance data — and it changes the interpretation entirely.

Simpson's paradox: the lift is a composition effect

The aggregate lift is a **Simpson's paradox**. Within strata, effects are small and **unstable across years**:

	2024			2025		
Stratum	**C**	**NC**	**Δ**	**C**	**NC**	**Δ**
Low	0.0	0.0	0	0.0	0.0	0
Border	7.5	4.7	+2.9	4.8	5.0	−0.2
Mid	50.0	54.7	−4.7	62.2	50.5	+11.8
High	94.9	94.8	+0.1	95.9	96.2	−0.3
Agg.	**42.9**	**32.2**	**+10.7**	**50.6**	**34.8**	**+15.8**

The positive signal **migrates** from borderline (2024) to mid (2025). If compliance were causal, the effect should be stable.

The key finding:

The aggregate lift (+10–16 pp) grows across years — but the within-stratum effects are noisy and inconsistent.

This is a **composition effect**: compliant and non-compliant papers are distributed differently across strata.

For low-scored papers: compliance is **futile**.

For high-scored papers: compliance is **unnecessary**.

The compliance incentive is a **mirage** that nonetheless drives research

Controlling for quality: matched pairs + regression

Matched-pairs design. Match each compliant borderline paper to a non-compliant one with the same rounded mean score. Compute within-pair acceptance difference.

Year	Pairs	Diff	p
ICLR 2024	964	+4.4 pp	0.016
ICLR 2023	504	+0.2 pp	1.000
Pooled	1468	+2.1 pp	0.160

After matching on score, the lift **largely disappears**. Pooled 95% CI: $[-0.8, +4.9]$ pp. Consistent with composition effect, not causal.

Proxy-control regression. Logistic regression predicting acceptance from compliance, controlling for mean score, number of reviewers, and review length.

Covariate	Coef (std.)
Compliance	-0.05
Mean score	$+0.92$
n reviewers	$+0.04$
Review length	-0.05

(Pooled borderline, $n = 6{,}359$)

Compliance coefficient is **near zero and negative** after controlling for quality. Review score dominates acceptance entirely.

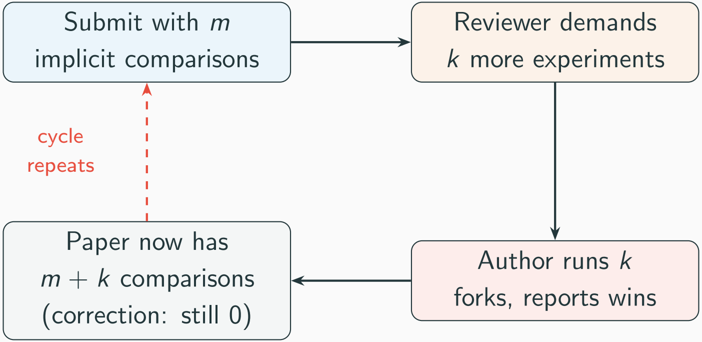
Why this matters: performativity driven by a mirage

The aggregate data tells researchers: "comply with experiment demands $\rightarrow$ get accepted." But the stratified data shows this is an illusion — compliance doesn't actually help within any score band. So what's happening?

1. Researchers *believe* more experiments = acceptance (the aggregate signal)
2. They optimize for compliance: run more benchmarks, add more ablations
3. This doesn't actually improve their acceptance odds (the stratified signal)
4. But it *does* inflate the total number of implicit comparisons
5. Which inflates the FWER — more false positives enter the literature

The review system reshapes research behavior based on a statistical artifact. Researchers game a signal that isn't real — and the gaming itself causes harm (more uncorrected comparisons). This is the counterperformative loop:

The counterperformative loop



Each cycle adds comparisons without adjusting the denominator. The FWER grows invisibly.

Part 4

What to do about it

Structural interventions

Make denominators visible

- Report total experiments run, not just those in the paper
- For SAEs: report interpretability *rates*, not cherry-picked features
- Template: "trained N features, k scored above threshold t"

Audit demands for FWER

- Before demanding an ablation, ask whether it increases the comparison count
- Demand the correction alongside the experiment

Publish negative results

- GDM's "research we didn't publish" post is the template
- Venues should create tracks for replications and null results

Bridge proof cultures

- Cross-community benchmarks (SAEBench, MIB are a start)
- Shared evaluation criteria across subfields
- Require engagement with alternative evidence standards in reviews

Project limitations

1. **Observational.** The demand-compliance analysis is correlational. Matched-pairs and proxy-control regression reduce but don't eliminate confounds (pooled matched-pairs: $p = 0.16$, CI crosses zero).
2. **One venue.** ICLR is the only major venue that publishes rejected submissions. Internal replication across 2023–2024 shows pattern stability.
3. **Effective, not Barnesian.** We claim benchmarks *reshape* research priorities, not that the world conforms to the benchmark in the strong Black-Scholes sense.
4. **No ethnography.** Citation insularity is statistically confirmed ($p = 0.0002$), but a proper sociology of MI would require interviews and lab observation.
5. **Infrastructure gap.** Even if norms changed, experiment-tracking tools default to private. The tooling for public denominator reporting doesn't exist yet.

Conclusion

Statistical reform treats the **symptom**.

The cause is an evaluation culture whose instruments both **measure** and **manufacture** what counts as knowledge.

The benchmark is an engine, not a camera.

And like all engines, it needs maintenance, inspection,

and occasionally — replacement.

Thank you

Paper: TBD

Repository: github.com/elliottower/benchmarks-engines

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Data: 11,196 ICLR submissions, 688-paper MI citation corpus, 18 manual paper audits, OpenAlex lifecycle analysis.

References i